

Effects of spatial structure on carbon emissions of urban agglomerations in China

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ABSTRACT

Urban agglomerations play a crucial role in carbon emissions as the primary spatial carriers of economic activities and population. Clarifying the relationship between the spatial structure of urban agglomerations and the per capita carbon emissions (PCCE) of their member cities is essential for formulating low-carbon growth strategies. However, the impact of polycentricity, a key characteristic of urban agglomeration spatial structure, on urban PCCE remains unclear. To address this, we selected 184 prefecture-level cities within 14 urban agglomerations across China, analyzing the evolving patterns of carbon emissions and spatial structures. Employing a series of fixed-effect models based on urban panel data from 2005 to 2022, we examined the effect and transmission mechanisms of polycentricity on PCCE. Our findings suggest a U-shaped relationship between the polycentricity of urban agglomerations and the PCCE of their member cities. Highly developed urban agglomerations are more likely to benefit from polycentric development. Enhancing regional transportation infrastructure and fostering technological innovation can help unlock the potential for carbon emission reductions associated with polycentric development.

1. Introduction

Urban areas are the primary contributors to carbon emissions, making their low-carbon transition essential for mitigating climate change (Chen et al., 2023). In 2020, urban areas accounted for an estimated 67–72 % of global greenhouse gas (GHG) emissions, a proportion expected to rise further by 2050 (Bai et al., 2022). In response to this growing challenge, governments worldwide have implemented urban mitigation strategies. For example, China has committed to peaking CO₂ emissions before 2030 and has proposed to promote the green and low-carbon transformation of urban development and the coordinated growth of urban clusters (The State Council of China, 2021). In this context, research on the impact of urban spatial structure on carbon emissions has attracted widespread attention (Cai, Ren, et al., 2023). Cities, with their strong interdependencies across scales, are embedded in broader ecological, economic, technological, and governance structures that often limit their systemic function (Bai et al., 2016). Examining carbon emissions and mitigation strategies at the regional urban system level is, therefore, essential. However, empirical studies on the relationship between urban agglomeration spatial structures and carbon emissions remain limited (Hong et al., 2022). It is urgent to explore the

mechanisms through which urban agglomeration spatial structures influence carbon emissions and to understand their heterogeneity.

Spatial structures play a crucial role in shaping carbon emissions within urban areas (Liu et al., 2020). The configuration of spatial structures—including transportation networks, industrial distribution, and energy flows—has profound implications for environmental pollution and carbon emissions (Wang & Li, 2023). Poorly planned spatial structures can inherently hinder sustainable development, exacerbating environmental and economic challenges (Wang, Niu, et al., 2022). The spatial structures of cities or urban agglomerations are generally classified as either monocentric or polycentric (Pan et al., 2024). Monocentric structures can generate positive externalities through enhanced energy efficiency, leading to significant reductions in carbon emissions (Han et al., 2018). However, the high-density concentration of population and employment in such structures may also produce negative externalities, such as traffic congestion, thereby increasing carbon emissions within the transportation sector (Cirilli & Veneri, 2014). In contrast, polycentric structures have the potential to mitigate transportation-related carbon emissions by fostering a more balanced distribution of residential and employment centers. Despite these theoretical insights, empirical research has yielded mixed findings

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regarding which spatial structure is more effective in reducing carbon emissions. Existing studies primarily focus on the urban and provincial scales, and case studies from different regions often present conflicting conclusions (Burgalassi & Luzzati, 2015; Sat, 2018; Yan & Li, 2023). Therefore, there is an urgent need to strengthen foundational research on carbon emissions within urban agglomerations to provide a robust theoretical framework and methodological support for developing emission reduction technological pathways.

As the world's largest carbon emitter, China faces substantial challenges in reducing carbon emissions. Urban agglomerations in China, which account for 77.8 % of the national GDP, generate 71.7 % of carbon emissions, and these figures are projected to rise to 90 % and 83 % by 2030, respectively (Rocky Mountain Institute, 2019). China has formed multiple urban agglomerations of different types under the government's development plan. Clarifying the impact of urban agglomeration spatial structure on the PCCE of member cities can provide theoretical guidance and scientific suggestions for controlling urban carbon emissions in China and other countries. In this context, this study aims to answer the following questions: (1) What are the characteristics of carbon emissions in China's urban agglomerations? (2) Does the polycentric spatial structure of urban agglomerations have better emission reduction performance? (3) What are the interactive mechanisms between the polycentricity of urban agglomerations and the PCCE of member cities?

The innovation and marginal contributions of this study are primarily reflected in three aspects: In terms of research perspective, previous studies have investigated the relationship between spatial structure and carbon emissions at the urban and provincial scales, but there are few studies at the urban agglomeration scale. Furthermore, most research of urban carbon emissions has focused on specific sectors, such as transportation and building, with fewer studies on a more comprehensive carbon emissions inventory. This paper contributes to this knowledge by investigating how the polycentricity of urban agglomerations influences the carbon emissions of member cities. We selected 184 member cities within 14 urban agglomerations across China as the sample, utilizing high spatiotemporal resolution carbon emission datasets and various urban data to describe the differentiated patterns of PCCE and polycentricity. This lays the foundation for enhancing the understanding of the relationship between spatial structure and carbon emissions in different types of urban agglomerations. In terms of research methodology, we applied a robust panel fixed-effects model, incorporating key control variables and testing for moderating effects to enhance the precision of estimating the impact of polycentricity on PCCE and its underlying mechanisms. In terms of research conclusions, the findings indicate that extremely high or low levels of polycentricity in urban agglomerations are not conducive to reducing the PCCE of member cities. Developed urban agglomerations are more likely to achieve carbon reduction through polycentric development. Additionally, regional transportation infrastructure and technological innovation can moderate the impact of spatial structure on carbon emissions. These results offer valuable insights for governments in formulating targeted spatial planning and urban mitigation policies for urban agglomerations.

2. Literature review

2.1. Spatial structure of urban agglomerations

The expansion of economic and social activities beyond urban boundaries drives the formation of urban agglomerations. The spatial structure of an urban agglomeration can be defined by the spatial layout and interrelationships among cities within the agglomeration, encompassing multiple dimensions, such as city size distribution and intercity connections (Pan et al., 2024). Various indicators, including urban primacy index, polycentricity, spatial compactness, transportation accessibility, and network connectivity, are commonly used to measure the

spatial structure of urban agglomerations (Liu et al., 2020; Zhang et al., 2020).

Polycentricity is a key feature of urban agglomerations, reflecting whether resources are concentrated in a single city or dispersed across multiple cities. This concept is useful for examining the size distribution of cities within an agglomeration (Phelps et al., 2023). The quantification of polycentricity primarily encompasses two aspects: morphological and functional. For morphological polycentricity, current research relies on socio-economic data, such as population, employment, GDP, and remote sensing data like nighttime lights. Methods employed to characterize the morphological polycentricity of urban agglomerations include the rank-size rule, Gini coefficient, primacy ratio, Pareto index, and Herfindahl-Hirschman index (Phelps et al., 2023; Zhuo & Liu, 2020). For functional polycentricity, studies often adopt a flow-space perspective, utilizing data related to transportation flows (e.g., highways and railways), capital flows (e.g., cross-regional investments), knowledge flows (e.g., patent collaborations), and location-based services (e.g., mobile signaling and Baidu Mobility data). The interactions among cities are typically assessed through methods like social network analysis to quantify the polycentricity of urban agglomeration networks (He et al., 2021; Zhu et al., 2021).

Urban agglomerations across different regions and developmental stages exhibit differentiated spatial structures. For instance, the Southeast of England and the Paris region are characterized by a single dominant core city, while Randstad and the Rhine-Ruhr exhibit polycentric characteristics (Hall, 2009). Urban agglomerations in China have moved away from a hierarchical structure centred on a dominant core city and now exhibit diverse development trend (Yu, Wei, et al., 2020). Extant studies have identified heterogeneous effects of urban agglomeration polycentricity on urban economic development and environmental conditions (Wang, Sun, & Zhang, 2022; Zhang & Zhang, 2022).

Based on the above analysis, we intend to conduct an empirical analysis of the relationship between urban agglomeration spatial structure and carbon emissions of member cities from the perspective of morphological polycentricity. There are two main reasons: firstly, theoretical insights suggest that polycentricity can serve as a bridge to link spatial structure with carbon emissions (Han et al., 2022; Sat, 2018). Polycentricity focuses on the distribution of different centres within cities, clearly reflecting the characteristics of city size distribution. According to theories of economies of scale and agglomeration economies, city size distribution relates to production efficiency, including the efficiency of energy and resource utilization (Tabuchi, 1998), which could therefore impact carbon emissions (Bai et al., 2022). Secondly, the polycentricity index based on the rank-size rule exhibits simplicity while also providing valuable information (Liu et al., 2020).

2.2. Carbon emissions of urban agglomerations

The extant literature on carbon emissions in urban agglomerations has provided valuable insights into their characteristics and the factors influencing them. Scholars have examined the spatiotemporal characteristics of carbon emissions, including total emissions (Qian et al., 2022), performance (Yu, Wu, et al., 2020), and spatial networks (Dong & Li, 2022), from perspectives such as the overall urban agglomeration and its member cities. Econometric models have been extensively employed to explore the drivers of carbon emissions (Wang & Li, 2023). The STIRPAT model, for instance, identifies population, affluence, and technology as the primary factors influencing environmental quality (Fang et al., 2022). Qian et al. (2022) utilized the Tapio decoupling analysis to examine the spatiotemporal relationship between carbon emissions and the impact of socioeconomic development in 19 urban agglomerations across China. Ahmad et al. (2021) found a unidirectional mixed causal linkage between urban agglomerations and the growth of carbon emissions.

Additionally, scholars have explored the factors influencing carbon emissions in urban agglomerations from a regional association

perspective, highlighting the spatial spillover effects of emissions. Yan and Li (2023) demonstrated that regional integration initiatives can significantly reduce urban carbon emissions, particularly when considering the expansion of regional integration. There is a notable impact on the spatial correlation of carbon emissions through geographical adjacency, population size, per capita GDP, and technology level. Spatial correlation and the spillover effects of carbon emissions increase among regions that share higher technological affinities (Wang, Rani, & Razzaq, 2023).

Although related studies have made some achievements in understanding the driving factors of urban agglomeration carbon emission, there are still some limitations to be addressed. Firstly, the existing literature primarily focuses on the relationship between socio-economic variables and carbon emissions, with little attention paid to structural factors within regional urban systems. Secondly, these studies focus on analyzing a specific urban agglomeration, ignoring the possibility of different development patterns between urban agglomerations. Cities within different urban agglomerations are influenced by their respective levels of economic development, energy consumption, industrial structures, and other factors, which result in varying levels of carbon emissions and differences in their ability to achieve carbon reduction (Shen et al., 2021). This difference requires policymakers to tailor carbon reduction policies for different cities based on the actual situation of each urban agglomeration.

2.3. Relationship between polycentric structure and carbon emissions

Scholars have varying views on the effectiveness of polycentric development as a policy tool for achieving carbon reduction. This uncertainty may arise from “the multi-scale nature” of the concept of polycentricity and “the multi-dimensional nature” of carbon emissions. Polycentricity can refer to at least three spatial scales: intra-urban, inter-urban, and national (Veneri & Burgalassi, 2012), and carbon emissions include sectors such as energy, industry, and transportation (Bai et al., 2022). With different scales and dimensions of polycentricity and carbon emissions, the results of existing research can be categorized into three types: positive correlation, negative correlation, and dual effects.

First, a polycentric structure promotes carbon emissions. A case study of 10 Chinese urban clusters demonstrated that the polycentric development of urban clusters promotes carbon emissions in the construction industry (Wang et al., 2025). For NUTS-3 statistical regions in Italy, a morphological polycentric structure slightly increased carbon emissions from residential heating and transportation, while a functional polycentric structure significantly increased carbon emissions from transportation (Burgalassi & Luzzati, 2015).

Second, a polycentric structure reduces carbon emissions. Rapidly developing cities with polycentric structures tend to have higher public transportation usage, thereby reducing transportation-related carbon emissions (Yang et al., 2020). Compared to the study based on Italian NUTS-3 statistical regions, another study based on NUTS-2 showed that both morphological and functional polycentric characteristics harm per capita energy use in the transportation sector (Veneri & Burgalassi, 2012). The difference between the two empirical results from Italy can be attributed to inconsistent spatial scales, as what looks polycentric at one spatial scale (e.g., NUTS-3) may be monocentric at higher scales (e.g., NUTS-2) (Burgalassi & Luzzati, 2015). In the Chinese provincial context, market segmentation has severely hindered China's energy efficiency, and polycentric development has a narrowing effect on regional energy efficiency disparities (Zou et al., 2019).

Third, when polycentric structures have either a positive or negative impact on carbon emissions, a dual effect may occur (Wang, Niu, et al., 2022). An empirical analysis based on 23 provinces of China indicated that the relationship between polycentric spatial structures of provincial urban systems and carbon emissions follows a U-shaped curve (Chen, Chen, & Song, 2021).

Overall, more evidence is needed to clarify the impact of polycentric

spatial structures at the urban agglomeration scale on broader carbon emissions. Most research on polycentric structures and carbon emissions focuses on the urban or provincial scale, with limited studies exploring the relationship at the urban agglomeration scale. They have mainly examined the relationship between polycentricity and carbon emissions in specific sectors, such as transportation. Furthermore, while much attention has been paid to spatial structure performance, fewer studies explore the internal mechanisms by which polycentric development affects carbon emissions. To achieve coordinated emission reduction in urban agglomerations, it is crucial to explore moderating factors to optimize polycentric development strategies.

To address these issues, the nonlinear impact of polycentricity of urban agglomerations on carbon emission of their member cities is the main content of this article. Then, we use a more comprehensive carbon emissions inventory data, calculating CO₂ emissions from fossil fuel combustion, cement production, and gas flaring. Finally, to explore the influential mechanism, the cross term of transportation infrastructure, technological innovation and polycentricity has been included in the regression model.

3. Research design

3.1. Research hypothesis

Polycentric and monocentric structures represent two distinct models of urban agglomeration development, each with unique characteristics that influence economic growth, resource allocation, and urban efficiency. Traditional agglomeration economic theory suggests that spatial concentration of factors can lead to cost savings and efficiency improvements, thereby driving economic growth (Tabuchi, 1998). In a monocentric structure, urban agglomerations develop around a primary central city, facilitating the concentration of economic factors and enabling agglomeration benefits through learning, sharing, and matching mechanisms (Eeckhout et al., 2014; Hörcher et al., 2020). However, this structure may also result in excessive resource concentration, leading to problems such as traffic congestion and environmental stress (Brinkman, 2016; Zhou et al., 2021).

When the urban agglomeration becomes highly polarized, resulting in agglomeration diseconomies, the polycentric structure reveals its advantages. This structure promotes a more balanced distribution of economic factors across the urban agglomeration, fostering coordinated development among cities and thereby enhancing the overall economic performance of the urban agglomeration (Pan et al., 2024). In a polycentric model, the agglomeration comprises multiple centres interconnected by transportation and communication networks, generating network externalities that extend beyond geographical boundaries (Huang et al., 2020). This structure helps mitigate the over-concentration issues of a dominant central city and enables agglomeration economies on a larger spatial scale by leveraging size effects and specialized collaboration (Cai, Ding, et al., 2023).

According to agglomeration economic theory, the polycentric development of urban agglomerations does not always guarantee the carbon emission performance of member cities in any situation. There are two main reasons: 1) The growing concentration of people and activities is an opportunity to increase resource efficiency and decarbonize at scale (Bai et al., 2022). When the centre is not large enough to induce any diseconomies, agglomeration economies occur in a single city centre. 2) Connection barriers among too many centres can compromise economic efficiency (Pan et al., 2024). Many studies have proposed nonlinear or threshold relationships between the spatial structure of urban agglomerations and factors such as green production efficiency and environmental performance (Chen, Zhang, & Ruan, 2021; Han et al., 2022). A U-shaped curve has been used for this characterization (Liu et al., 2022). Therefore, based on the above analysis, the first hypothesis is as follows:

H1. The relationship between the polycentricity of urban agglomerations and the PCCE of member cities follows a U-shaped curve.

Polycentric development is often advocated as a strategy to mitigate the diseconomies of agglomeration. In the context of urban agglomeration development in China, governments play a crucial role in shaping spatial structures by regulating key development indicators across cities within an agglomeration. Through policy interventions, governments designate central and member cities, influencing the allocation, flow, and concentration of resources among them (Li et al., 2022). For instance, the Central Plains Urban Agglomeration in China has implemented a development strategy aimed at strengthening the central city, thereby enhancing its agglomeration efficiency. However, this strategy has also resulted in an excessive concentration of development resources (Yang et al., 2022). As urban agglomerations reach a certain level of economic development, the overcrowding of central cities may reduce overall efficiency and hinder the low-carbon transition of member cities. Therefore, the second hypothesis of this study is as follows:

H2. Urban agglomerations with high levels of development are more likely to benefit from polycentric development in terms of carbon reduction.

Borrowing size effect in polycentric urban agglomerations can be strengthened with the improvement of regional transportation facilities (Wang, Sun, & Zhang, 2022). A specialized division of labour means that each region carries out specialized production based on its comparative advantages, thereby achieving economies of scale and improving overall economic and energy efficiency (Volgmann & Rusche, 2020). The

improvement of transportation infrastructure significantly reduces regional transportation costs and time, which may trigger industrial restructuring and promote the formation of a polycentric spatial structure (Zhang et al., 2020), leading to an improvement in urban carbon emission efficiency (Li & Cheng, 2022). Therefore, the third hypothesis is as follows:

H3. Improving regional transportation infrastructure is conducive to the release of carbon emission reduction potential in polycentric development.

Polycentric urban agglomerations have a diffusion effect of knowledge spillover and innovation upgrading, which promotes technological interaction between central cities and noncentral cities (Solis et al., 2022). Central cities leverage their advantages in gathering innovative elements such as talent and knowledge to promote the agglomeration of high-tech industries, which can indirectly reduce carbon emissions by improving innovation levels (Wang, Dong, & Dong, 2022). Additionally, enterprises can obtain knowledge spillover from other cities through geographically dispersed units (Mao et al., 2023). With the dispersed layout of enterprises and high-skilled labour flow, non-central cities gradually absorb advanced technology and experience from centre cities, and achieve an overall improvement in innovation efficiency and technical level of resource utilization (Wang, Li, & Liao, 2022; Wang, Wang, et al., 2022).

H4. Stimulating technological innovation is conducive to the release of carbon emission reduction potential in polycentric development.

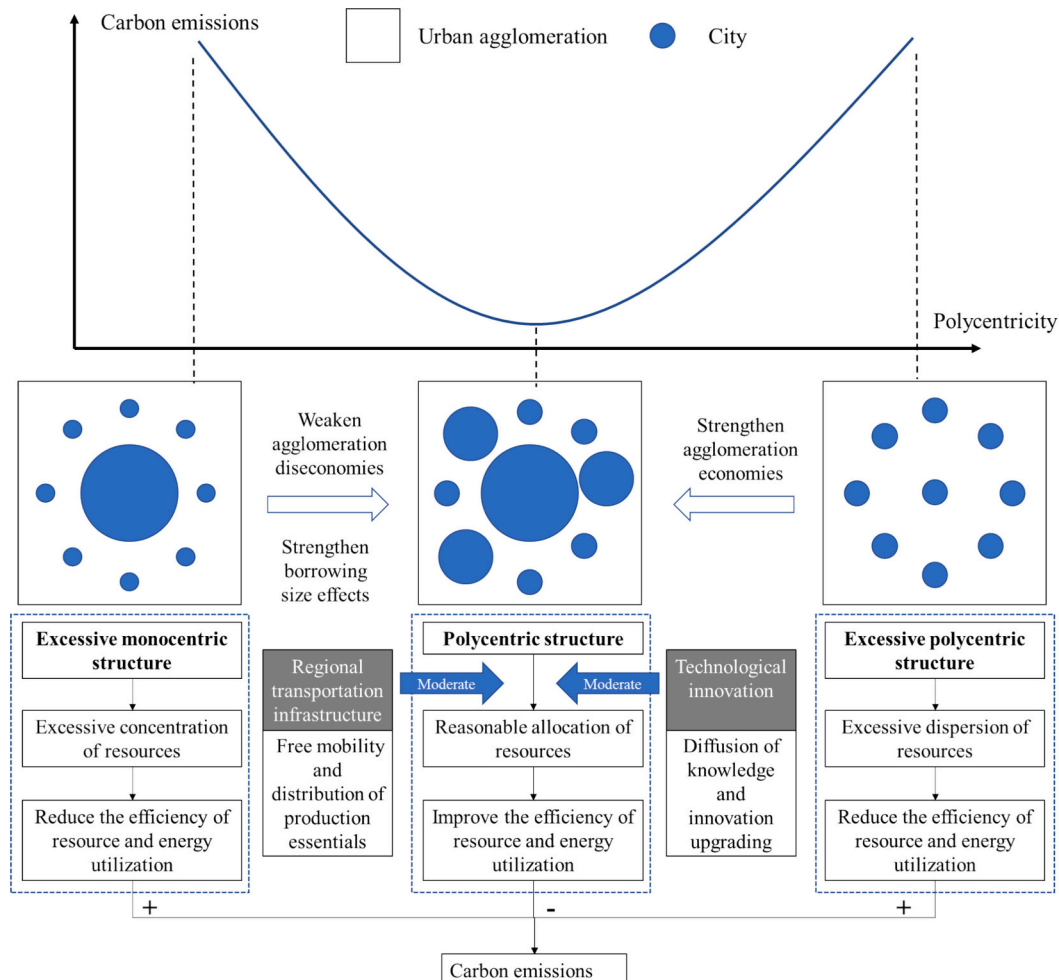


Fig. 1. The mechanism of polycentric structure affecting carbon emissions.

Based on previous studies, the polycentricity of urban agglomerations may influence the carbon emissions of member cities via the following mechanisms (see Fig. 1).

3.2. Study area

The 14th Five-Year Plan proposed by China's government identified 19 major urban agglomerations. We selected 184 prefecture-level member cities from 14 out of the 19 major urban agglomerations as our research subjects (Fig. 2). On the one hand, for the five urban agglomerations not included, key datasets were incomplete and lacked consistency with the remaining 14. To ensure a robust and comparable analysis across all selected regions, we chose to focus on areas where comprehensive and consistent data was available. On the other hand, the selected 14 urban agglomerations represent a significant proportion of the population, economic activity, and urbanization trends in China. These regions are of particular interest due to their distinctive development patterns and policy relevance, aligning well with the objectives of our study. We identify the boundaries of each urban agglomeration by consulting the development plan of urban agglomerations announced by the National Development and Reform Commission. For those that have not been announced, such as the Central and Southern Liaoning urban agglomeration (CSL), we referred to the boundaries delineated in the China Urban Agglomeration Development Report (Fang et al., 2016).

According to the 14th Five-Year Plan, these agglomerations represent three typical types of urban agglomerations with different development levels. Firstly, the Beijing-Tianjin-Hebei region (BTH), the Yangtze River Delta (YRD), the Pearl River Delta (PRD), the Chengdu-Chongqing region (CC), and the middle reaches of the Yangtze River (MY) are optimizing and upgrading urban agglomerations (OUUA) with good connectivity and the highest development level. Secondly, five urban agglomerations of the Shandong Peninsula (SP), the coastal areas of Guangdong, Fujian, and Zhejiang (GFZ), the Central Plains (CP), the Guanzhong Plain (GP), and the Beibu Gulf (BG) are developing and strengthening urban agglomerations (DSUA). The economic development of DSUA is swift, but the industrial correlation between cities

needs enhancement. Thirdly, the four urban agglomerations of Harbin-Changchun (HC), Central and Southern Liaoning (CSL), Central Shanxi (CS), and Huhhot-Baotou-Ordos-Yulin (HBOY) are nurturing urban agglomerations (NUA) with the lowest development level. There is a weak coordinated development of industries between non-central cities and central cities in NUA. We supplement the theoretical evidence of the relationship between spatial structure and carbon emissions and propose targeted policy suggestions for different urban agglomerations by examining the heterogeneity of the three types of urban agglomerations.

3.3. Methods

3.3.1. Exploratory spatiotemporal data analysis

Exploratory spatiotemporal data analysis (ESTDA) is instrumental in describing the spatiotemporal distribution patterns of data. In this study, the spatiotemporal dynamics of per capita carbon emissions (PCCE) in member cities of urban agglomerations are examined using local spatial autocorrelation and spatiotemporal transition analysis, based on the local indicator of spatial association (LISA). Spatiotemporal transition theory provides the theoretical foundation for constructing the transition matrix (Rey & Janikas, 2009).

As shown in Table 1, spatiotemporal transition types are classified into four categories: Type I, Type II, Type III, and Type IV. These categories represent different transition patterns of local and neighboring PCCE:

Exploratory spatiotemporal data analysis is instrumental in describing the spatiotemporal distribution patterns of data. In this study, the spatiotemporal dynamics of PCCE of member cities in urban agglomerations are analyzed employing local spatial autocorrelation and spatiotemporal transition, based on the local indicator of spatial association (LISA). Spatiotemporal transition theory lays the foundation of the transition matrix (Rey & Janikas, 2009). As shown in Table 1, the spatiotemporal transition types are classified into four categories: Type I (both local and neighboring PCCE modes remain stable), Type II (only local PCCE mode undergoes transition), Type III (only neighboring PCCE mode undergoes transition), and Type IV (both local and neighboring

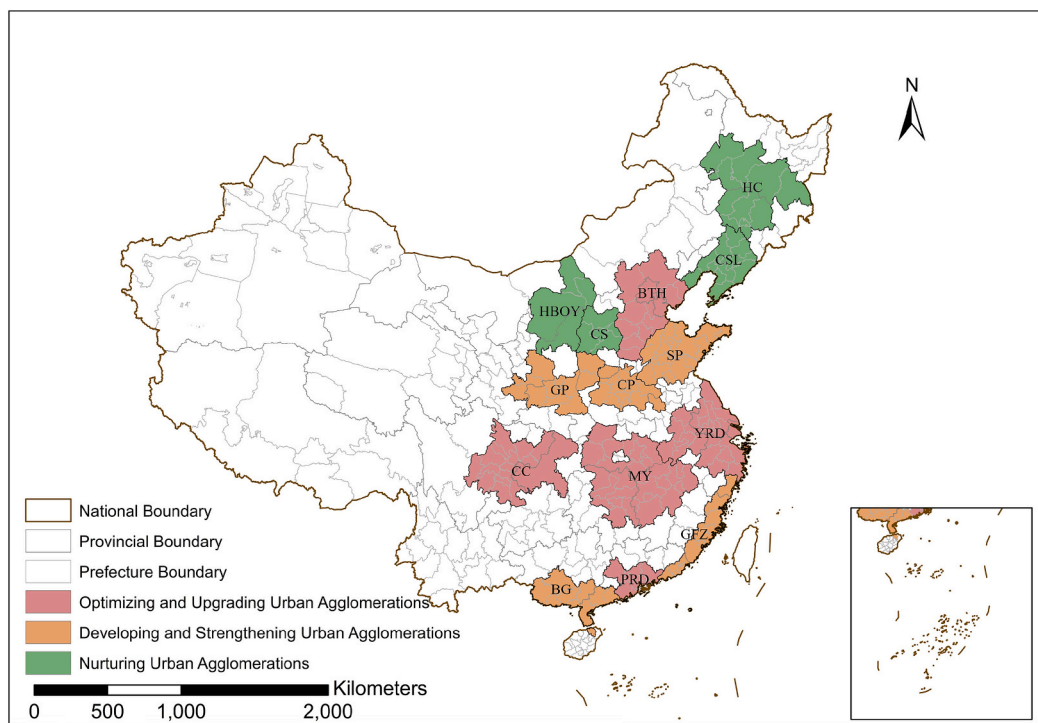


Fig. 2. 14 Urban agglomerations in China.

Table 1

Basic types of spatiotemporal transition.

Type	Spatiotemporal transition form	Symbol
Type I	Both local and neighboring PCCE modes remain stable	$HH_t \rightarrow HH_t, LL_t \rightarrow LL_t, HL_t \rightarrow HL_t, LH_t \rightarrow LH_t$
Type II	Only the local PCCE mode undergoes transition	$HH_t \rightarrow LH_t, HL_t \rightarrow LL_t, LL_t \rightarrow HH_t$
Type III	Only the neighboring PCCE mode undergoes transition	$HH_t \rightarrow HH_t, HL_t \rightarrow HH_t, LH_t \rightarrow LL_t, LL_t \rightarrow LH_t$
Type IV	Both local and neighboring PCCE modes undergo transition	$HH_t \rightarrow LL_t, HL_t \rightarrow LH_t, LH_t \rightarrow HL_t, LL_t \rightarrow HH_t$

Remarks: HH_t represents that the PCCE of a prefecture-level city in year t is characterized by High High Cluster (HH) modes, meaning that the city has high PCCE, and its surrounding areas also have high emissions. Similarly, HL_t means that the city has high PCCE in year t' while its surrounding areas have low emissions. Comparing different years, when a prefecture-level city converts from HH to HL, this transition is classified as Type III.

PCCE modes undergo transition). Type IV can be subdivided into Type IVA and Type IVB, representing the situation of co-migration and reverse migration of local and neighboring PCCE, respectively.

According to the transition type of PCCE, space-time cohesion (SC) can be calculated as follows (Ma et al., 2022):

$$SC = \frac{\text{Type I} + \text{Type IVA}}{n} \quad (1)$$

Where Type I and Type IVA represent the transition number of Type I and Type IVA; n represents the total number of transitions. The higher the SC, the stronger the spatial locking effect.

3.3.2. Regression model

The nexus between the spatial structure of urban agglomerations and the carbon emissions of member cities can be comprehended by utilizing a fixed effect model based on panel data. In our study, the Hausman test suggested that fixed-effect models should be preferred. The fixed-effect model can control for factors that do not change over time at the individual level and factors that do not change with individuals in the same year, thus enabling a more accurate estimation of the impact of explanatory variables. The PCCE of member cities and the polycentricity in each urban agglomeration are examined as the explained variable and as the explanatory variable, respectively. Specifically, PCCE was converted into a logarithmic form to eliminate heteroscedasticity. The baseline econometric model is as follows:

$$\ln PCCE_{ijt} = \beta_0 + \beta_1 Q_{jt} + \beta_2 X_{ijt} + \mu_{ij} + \varphi_t + \varepsilon_{ijt} \quad (2)$$

where i represents the member cities of the urban agglomeration j ; t represents the year. β_1 is the correlation coefficient; β_0 is the intercept term; μ_{ij} is the individual fixed effect; φ_t is the time fixed effect, and ε_{ijt} is the error term. $PCCE_{ijt}$ denotes the per capita annual carbon emission of the city i within the urban agglomeration j . Q_{jt} represents the polycentricity of the urban agglomeration j where city i is located. X_{ijt} denotes control variables.

To test Hypothesis 1, we added the quadratic term of Q to the baseline model as follows:

$$\ln PCCE_{ijt} = \beta_0 + \beta_1 Q_{jt} + \beta_2 Q_{jt}^2 + \beta_3 X_{ijt} + \mu_{ij} + \varphi_t + \varepsilon_{ijt} \quad (3)$$

To test Hypothesis 2, we conducted grouped regression analyses based on model (2), categorizing urban agglomerations into three types: OUUA, DSUA, and NUA. To further verify Hypotheses 3 and 4, we incorporated mechanism variables and introduced an interaction term between Q and the mechanism variables.

3.3.3. Selection of variables

We selected per capita carbon emissions (PCCE) as the explained variable of this study. It is believed that the total carbon emissions of a

region are determined by geographical location, factor endowments, and historical factors, while the efficiency of carbon emissions is a comprehensive manifestation of urban economic decision-making and environmental behaviour (Fang et al., 2022). Extant research revealed that burning fossil fuels to generate energy for buildings, transportation, industry, and other sectors is a major source of urban GHG emissions (Gurney et al., 2015). Meanwhile, physical-chemical reactions in the industrial process, like cement production, are also essential factors in carbon emissions. As a commonly used method in literature, fossil CO2 emissions are usually based on energy statistics and cement production data (Friedlingstein et al., 2022; Liu et al., 2015). In this study, carbon emission data originated from the Open-Data Inventory for Anthropogenic Carbon Dioxide (ODIAC), which calculates CO2 emissions from fossil fuel combustion, cement production, and gas flaring by combining a worldwide point source database and satellite observations of the global nighttime distribution. Because of the inherent characteristics of the nighttime distribution, source regions corresponding to urban agglomerations were well articulated. With a temporal resolution of months and a spatial resolution of 1 km, it is a global dataset of CO2 emissions (Oda et al., 2018). We added the monthly CO2 emissions to get the annual emissions of cities and then divided them by the city's permanent population to obtain PCCE. The calculation formula is as follows:

$$PCCE_i = \frac{\sum CO2}{\text{population}_i} \quad (4)$$

Polycentricity was used to measure urban agglomeration spatial structure, which was the explanatory variable. Based on the rank-size rule, the polycentricity of urban agglomerations is measured per the population and economic distribution in the region (Meijers & Burger, 2010). In general, there is a high positive correlation between the size of the urban population and the economy. We adopted an economy-centered measurement approach. The specific calculation formula is as follows:

$$N_i = \frac{N_i}{R_{Ni}^{qn}} \quad (5)$$

Convert to logarithmic equation form:

$$\ln N_i = \ln N_i - Q \ln R_{Ni} \quad (6)$$

where N_i denotes the GDP of the city i . The GDP of cities in the urban agglomeration are sorted in order of decreasing size, with ranks represented by R_{Ni} . Q are the parameters to be estimated, representing the polycentric index. Q measures the polycentric spatial structure of urban agglomeration, which is a negative indicator. When $Q > 1$, the distribution of economy in the urban agglomeration tends to be monocentric. When $Q < 1$, the distribution of economy in the urban agglomeration is relatively balanced, showing a polycentric structure. The smaller the Q value, the more evident the development trend of polycentricity.

The study controls for the following variables. (1) Economy (ECO), population (POP), and technological innovation (TECH): The Stochastic Impacts by Regression on Population, Affluence, and Technology (STIRPAT) model has been widely employed in social science research to evaluate the impact of human activities on the environment, which empirically demonstrated the correlation between ECO, POP, TECH and carbon emissions (Yu, Wu, et al., 2020). Specifically, TECH is also the mechanism variable in our study. (2) Industrial structure (IND): It's believed that the tertiary industry generally generates less carbon emissions than the secondary industry (Li et al., 2022). (3) Government intervention (GOV): Local government fiscal expenditure and its allocation affect regional economic and social development (Li & Cheng, 2022). (4) Foreign investment (FDI): Based on the pollution haven hypothesis, FDI degrades the quality of the environment. Due to the laxness in environmental regulations, polluting industries from developed economies are encouraged to set up business operations in developing

nations, thus providing a pollution haven for the company (Akar, 2019). (5) Infrastructure (INF): Transportation infrastructure increases urban carbon emissions and intensity (Xie et al., 2017). (6) Natural factors, including heating degree days (HDD) and cooling degree days (CDD): Climate change and global warming affect the amount of energy required and the structure of this demand, thereby influencing the amount of carbon emissions produced. A degree-day is a truncation of the daily temperature series at a base temperature, which can be used to estimate related energy demand under different climate scenarios (Hao et al., 2016). For other factors that do not change over time, such as geographical location, fixed effects can eliminate the interference of these factors on the regression results.

We selected regional transportation infrastructure (RI) and TECH as the two mechanism variables. They were used as moderators to observe the reduction effect of polycentric development of urban agglomerations on PCCE under different scenarios. RI is a dummy variable that takes on a value of 1 if member cities have high-speed railway stations and 0 otherwise. Table 2 contains the detailed information for each variable.

3.4. Data source

Considering the availability and integrity of sample data, the data selected in this empirical study are the panel data of 184 cities in China's urban agglomerations from 2005 to 2022. Carbon emissions data originates from ODIAC. The variables at the urban level, including GDP, permanent population, added value of industry, and so on, are from the China Urban Statistical Yearbook and CEIC Statistics Database. High-speed rail information for Chinese cities is available on the websites and public data of the National Railways Administration and the China Railways Corporation. HDD and CDD are calculated using the daily dataset V3.0 of ground climate data in China released by the China Meteorological Data Network.

4. Results

4.1. Spatiotemporal characteristics of carbon emissions in urban agglomerations

We aggregated the carbon emissions of 184 prefecture-level cities in 14 urban agglomerations and plotted a monthly heatmap (Fig. 3). It can be observed that from 2005 to 2022, monthly carbon emissions generally showed an upward trend. In terms of seasonal variation, carbon emissions peak in summer and winter. This may be because summer and winter are seasons with extreme temperatures: high temperatures in summer lead to a significant increase in the use of cooling equipment, while the cold in winter creates substantial heating demand. These factors significantly increase energy demand for residential life and

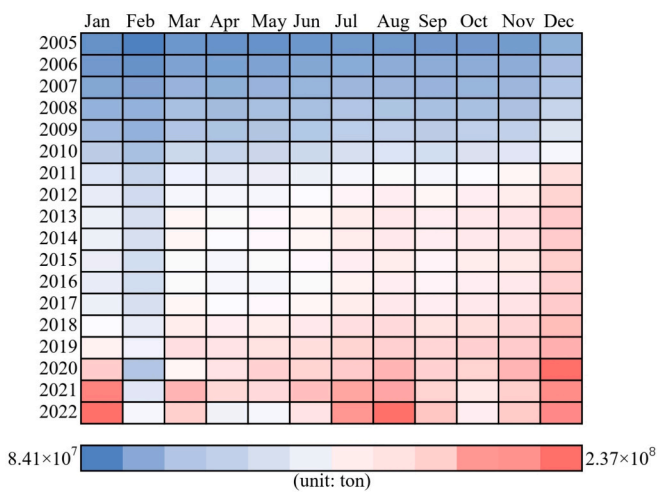


Fig. 3. Monthly heatmap of PCCE in China's urban agglomerations from 2005 to 2022.

industrial production, thereby raising energy-related carbon emissions.

We visualized the distribution of PCCE in prefecture-level cities and classified them according to the natural breaks method (Fig. 4). The results indicate an overall increasing trend in PCCE, with a clear spatial pattern: higher PCCE in the north and lower PCCE in the south. Among urban agglomerations, HBOY exhibited the highest PCCE, followed by CS, HC, YRD, CSL, BTH, and PRD. The elevated emissions in HBOY and CS can be attributed to their abundant fossil fuel resources, as cities with coal mines and high-intensity industries tend to generate greater carbon emissions (Yang et al., 2021). In contrast, the high PCCE observed in YRD, BTH, and PRD is likely due to higher economic development levels, which result in greater electricity, water, and gas consumption per capita. Additionally, heating in winter was required due to the high latitude of northern regions. In the northeastern region, the dual impact of high latitude (requiring heating) and population contraction may explain the elevated PCCE in HC and CSL. The urban agglomerations with the lowest PCCE of cities were CC and MY, followed by BG, CP, and GFZ.

As shown in Fig. 5, the total carbon emissions and PCCE of urban agglomerations experienced a notable expansion between 2005 and 2012. However, a sudden decline in 2013 coincides with the stricter air pollution control policies implemented that year. In November 2012, China convened the 18th National Congress of the Communist Party. The concept of ecological civilization was formally integrated into the country's development strategy, establishing "Beautiful China" as a key objective. This initiative promoted green and low-carbon development,

Table 2
Variable definitions.

Category	Variable	Description	Unit	Mean	SD	Min	Max
Explained variable	lnPCCE	Per capita carbon emission of member cities	ton per person	0.480	0.718	-2.567	2.757
	Q	Polycentricity of urban agglomerations	/	0.826	0.252	0.406	1.422
Control variables	ECO	Per capita real GDP	yuan per person	3.725	2.548	0.357	16.050
	POP	Population density	1000 people per km ²	0.611	0.699	0.017	8.852
	IND	The ratio of the added value of the secondary industry to that of the tertiary industry	ratio	0.944	0.507	0.129	5.348
	GOV	Proportion of general public budget expenditure to GDP	ratio	0.159	0.072	0.044	0.656
	FDI	Proportion of actually utilized foreign direct investment to GDP	ratio	0.003	0.003	2e-07	0.028
	INF	Per capita area of roads	m ² per person	4.634	3.498	0.221	32.143
	HDD	Heating degree days	10 ³ °C	1.126	1.211	0	5.551
	CDD	Cooling degree days	10 ³ °C	0.516	0.475	0	2.076
	TECH	Per capita authorized invention patents	piece per 10 ⁴ person	1.360	2.810	0	40.346
	RI	Whether member cities have high-speed railway stations	yes =1; no =0	0.594	0.491	0	1

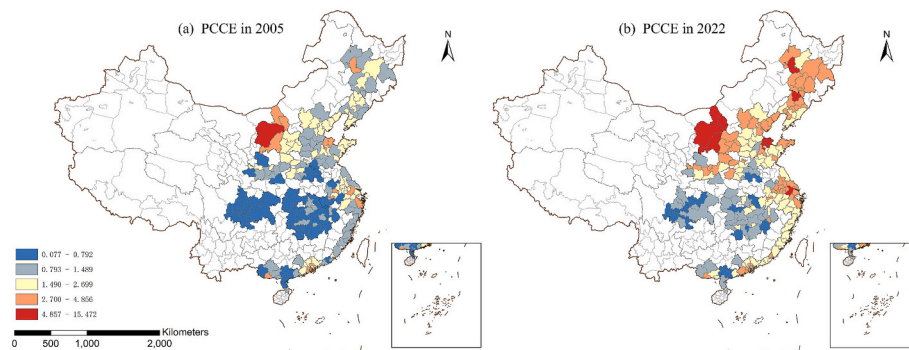


Fig. 4. Spatiotemporal characteristics of PCCE in China's urban agglomerations in 2005 and 2022.

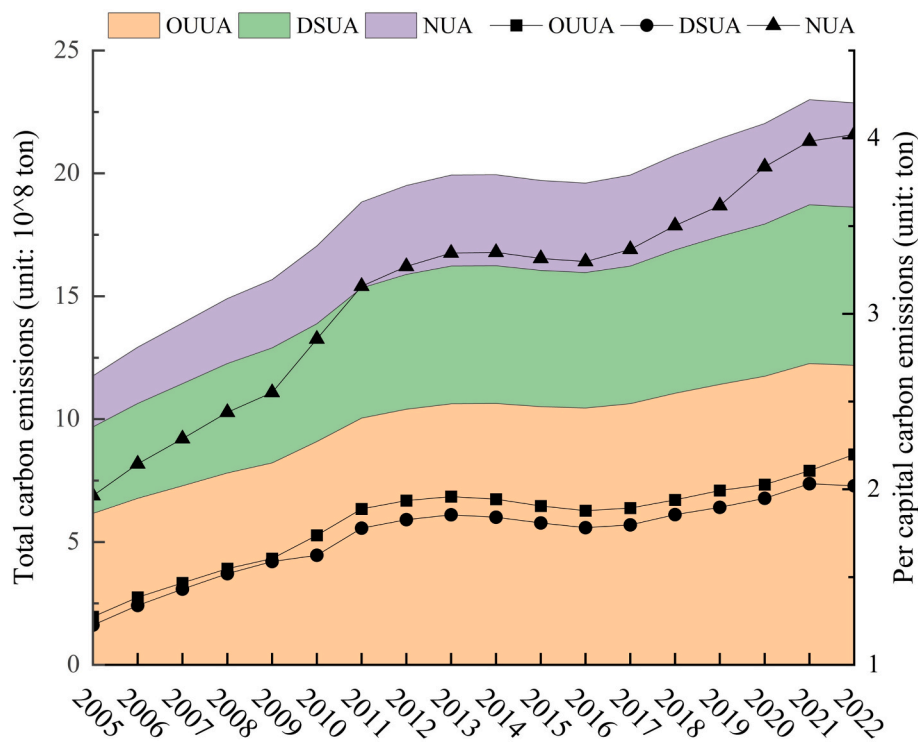


Fig. 5. The total carbon emissions and PCCE of three types of urban agglomerations from 2005 to 2022.

emphasizing the collaborative control of multiple pollutants and strengthened environmental supervision. These policies facilitated the elimination of outdated production capacity and the advancement of cleaner production technologies, contributing to the temporary reduction in carbon emissions. However, since 2016, carbon emissions have resumed their upward trend, highlighting the ongoing challenges in achieving long-term emission reductions. Regarding urban agglomerations with different development levels, OUUA had the highest carbon emissions, as it includes BTH, YRD, and PRD, where intensive economic activity and high population density drive substantial energy consumption. The total carbon emissions of DSUA were comparable to those of NUA, but PCCE in NUA was significantly higher than in both OUUA and DSUA.

Prefecture-level units within urban agglomerations mainly show Low-Low Cluster (LL) and High-High Cluster (HH) modes of PCCE (Fig. 6). LL units are mostly located in CC, MY, BG, and GFZ. The number of LL units has increased from 53 in 2005 to 60 in 2022. HH units are mainly located in CS, HBEY, and CSL. The number of HH units decreased significantly from 37 in 2005 to 30 in 2022, and the distribution centre was gradually relocated to the northwest. The disappearance of the LH

concentration area around BTH is related to the vigorous promotion of the industrial policy, which encouraged small and medium-sized state-owned enterprises to withdraw from the secondary industry and engage in the tertiary industry. To illustrate, since 2014, BTH has been promoting the relocation of non-capital functions from Beijing. In Beijing, a special action called the “Relocation, Remediation, and Promotion” campaign has been launched, resulting in the closure of nearly 3000 general manufacturing and polluting enterprises.

The spatial structure of PCCE across regions remained relatively stable over time. The series of global Moran's I showed a change in 2011; thus, the spatiotemporal transition method was used to further investigate the evolution characteristics of PCCE in two periods (2005–2011 and 2011–2022). The results are shown in Table 3. During two research periods, most of the local Moran's I scatter points remained in the same quadrant (Type I). The transition probabilities of the other types were low. The transition frequency of Type II and Type III from 2005 to 2011 was 8 and 3 times, respectively, while the transition frequency of Type II and Type III from 2011 to 2022 was 3 times. According to formulas (6), the spatiotemporal cohesion (SC) in the two periods was 0.860 and 0.925, respectively, indicating that the probability of no spatiotemporal

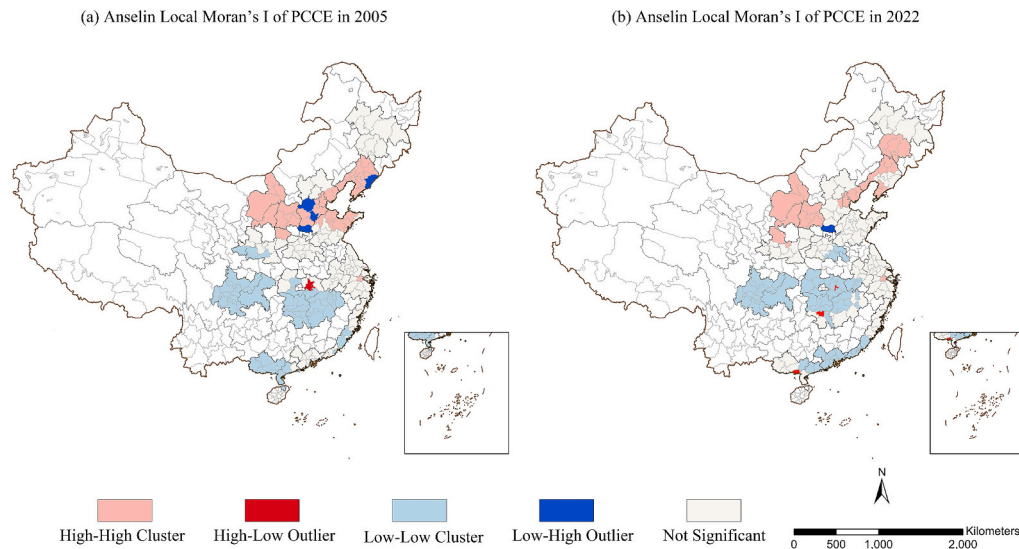


Fig. 6. Anselin Local Moran's I of PCCE in 2005 and 2022.

Table 3
Spatiotemporal transition matrices of local Moran's I.

period	t/t'	HH _{t'}	HL _{t'}	LH _{t'}	LL _{t'}
2005–2011	HH _t	I (n = 32)	III(n = 1)	II (n = 3)	IV(n = 1)
	HL _t	III (n = 0)	I (n = 1)	IV(n = 0)	II (n = 1)
	LH _t	II (n = 0)	IV(n = 0)	I (n = 3)	III (n = 1)
	LL _t	IV(n = 0)	II (n = 4)	III(n = 1)	I (n = 47)
2011–2022	HH _t	I (n = 22)	III(n = 0)	II (n = 1)	IV(n = 6)
	HL _t	III (n = 0)	I (n = 0)	IV(n = 1)	II (n = 1)
	LH _t	II (n = 0)	IV(n = 0)	I (n = 0)	III (n = 2)
	LL _t	IV(n = 8)	II (n = 1)	III (n = 1)	I (n = 51)

transition is 86.0 % and 92.5 %. The spatial agglomeration of PCCE had strong path dependence and locking characteristics. Past policies and planning decisions have had long-term impacts on carbon emissions within urban agglomerations. For instance, previous investments in transportation infrastructure and land use policies could influence the choice of transportation modes and urban layouts. This development inertia suggests that historical policy choices may continue to shape

carbon emissions in the future, potentially leading to either the persistence of high-emission zones or sustained reductions in specific areas.

4.2. Polycentricity of urban agglomerations

Most of China's Urban agglomerations presented a polycentric pattern. Fig. 7 illustrates the polycentricity of 14 urban agglomerations from 2005 to 2022, indicating a range of values for the polycentricity index from a minimum of 0.41 to a maximum of 1.42. When the polycentricity index is <1, it signifies that the urban agglomeration has a polycentric spatial structure. Among the 14 urban agglomerations, 11 demonstrated a polycentric structure, whereas the remaining 3 exhibited a monocentric structure.

Concerning polycentric urban agglomerations, between 2005 and 2022, there was evident reinforcement of the polycentric spatial structure in some regions like YRD and BG. In polycentric regions, regional planning policies that were aimed at promoting balanced growth were of paramount importance. Probably, investment in infrastructure connecting multiple centres and incentives for businesses to disperse symmetrically across the region contributed to this polycentric

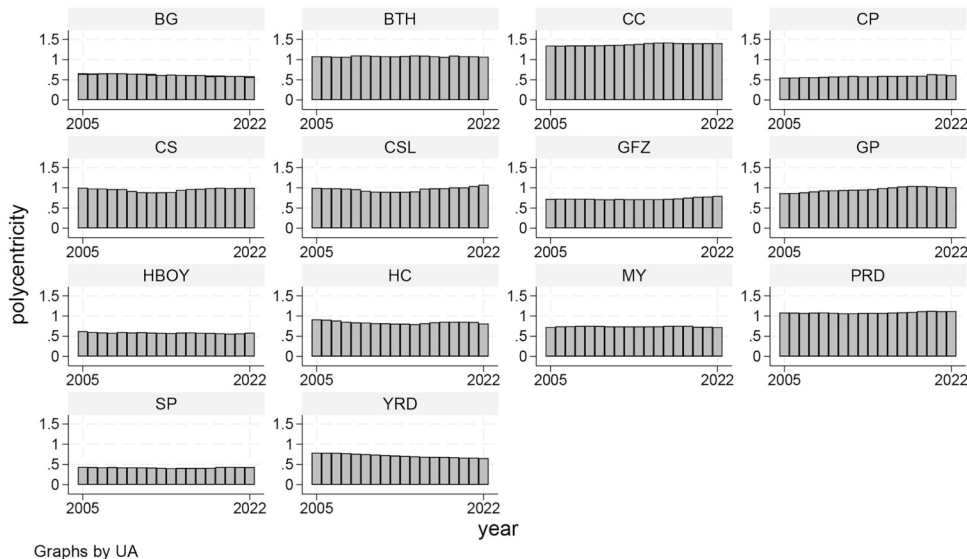


Fig. 7. Polycentricity of 14 Urban Agglomerations in China from 2005 to 2022.

strengthening. For example, in 2016, the State Council approved the “Yangtze River Delta Urban Agglomeration Development Plan,” which proposed to cultivate multi-level and diversified development axes in the YRD based on the transportation network. The plan emphasized promoting the development of the Nanjing metropolitan area, Hangzhou metropolitan area, Hefei metropolitan area, Suzhou-Wuxi-Changzhou metropolitan area, and Ningbo metropolitan area. These regions fostered balanced regional development by creating interconnected independent urban centres.

Regarding monocentric urban agglomerations, there has been a reinforcement of the monocentric structure of CC, BTH, and PRD. This shows that within these agglomerations, a primary city had continual dominance over economic activity and population concentration. The central city relied on improving infrastructure, public service conditions, and employment opportunities to continuously attract resources and population, leading to the strengthening of the monocentric structure.

Additionally, the polycentricity of some urban agglomerations like SP exhibited fluctuations and nonlinear changes. The possible reason is that major events, such as administrative division adjustments, changed the allocation of resources between cities and the development space of central cities. For instance, SP is a relatively evenly distributed urban agglomeration, lacking attractive mega cities like Beijing, Shanghai, Guangzhou, and Shenzhen. Jinan and Qingdao are the central cities of this system, but their radiating and driving capabilities are not strong (Liu et al., 2020). From 2005 to 2018, the polycentricity of SP gradually decreased from 0.436 to 0.415. However, in 2019, the State Council approved the revocation of Laiwu City in SP and transferred its jurisdiction to Jinan. Jinan greatly expanded its economy and population by merging with Laiwu, enhancing the strength of the provincial capital and laying the foundation for leveraging economies of scale. Under the circumstances, the polycentricity of SP sharply rebounded to 0.434 in 2019 and remained around 0.43 in recent years.

Table 4
The impact of polycentricity on PCCE.

Variable	I	II	III
Q	0.180 (0.112)	0.257*** (0.067)	-2.755*** (0.323)
ECO		0.070*** (0.002)	0.069*** (0.002)
POP		-0.212*** (0.016)	-0.207*** (0.015)
IND		-0.032*** (0.009)	-0.033*** (0.009)
FDI		4.444*** (1.102)	5.410*** (1.091)
GOV		1.747*** (0.058)	1.726*** (0.057)
INF		0.015*** (0.002)	0.012** (0.002)
TECH		-0.014*** (0.001)	-0.016*** (0.001)
HDD		0.016** (0.006)	0.017** (0.006)
CDD		0.041*** (0.010)	0.046*** (0.010)
Q2			1.716*** (0.180)
_cons	0.332*** (0.093)	-0.158** (0.057)	1.092*** (0.140)
Individual fixed effect/time fixed effect	YES	YES	YES
R ²	0.001	0.674	0.683
N	3312	3312	3312

Remarks: Standard error value is in (); * passed the 10 % significance test; ** passed the 5 % significance test; *** passed the 1 % significance test.

4.3. The impact of polycentricity on carbon emissions

Table 4 shows the overall estimation for the impacts of the polycentricity of urban agglomerations on the PCCE of member cities. As exemplified in Column III, the coefficient of Q is significantly negative and its square term is significantly positive at the statistical level of 1 %, indicating that there is a U-shaped relationship between polycentricity and PCCE (Fig. 8). On the left side of the inflexion point, PCCE decrease when the Q value increases, indicating that monocentric development is beneficial to carbon emission reduction. Where the inflexion point is exceeded, carbon emissions increase as the Q value increases, specifying that polycentric development is more conducive to carbon emission reduction. Excessive concentration in a single centre or absolute polycentric development with equal resources is unconducive to reducing carbon emissions, verifying Hypothesis 1.

In terms of control variables, the coefficients of ECO, FDI, GOV, INF, HDD, and CDD are significantly positive. (1) Economic growth has a positive impact on carbon emissions. This is because energy is the support of economic development. As an important input for industrial production and residents' lives, the consumption of fossil fuels accompanying economic development will lead to an increase in PCCE. (2) FDI has a positive impact on urban carbon emissions, which is consistent with the pollution haven hypothesis. Although some studies have shown that FDI can bring advanced technology through technology spillovers, the technological efficiency improvement brought by FDI is weakened by the influx of foreign energy consumption and high-polluting industries (Yu, Wu, et al., 2020). (3) For the current governance of China's urban agglomerations, government spending may suppress corporate environmental investment and green innovation through the “crowding out effect” (Hunt & Fund, 2016), thereby dragging down carbon emission control. (4) Cities with higher per capita road area have higher PCCE. A larger per capita road area may mean that cities rely more on the use of private cars, as there are more road resources available for use (Gallego et al., 2013). Private cars typically have higher carbon emissions compared to public transportation. (5) A rise in HDD and CDD increases the energy demand for winter heating and summer cooling, thereby increasing per capita energy consumption and carbon emissions (Hao et al., 2016).

The coefficient of POP, IND, and TECH are significantly negative, indicating that: (1) under the same conditions, cities with higher population density have lower PCCE. The possible reason is that the concentration of population and activities is conducive to improving resource utilization and public service efficiency (Bai et al., 2022). (2) The higher the ratio of tertiary industry-added value to secondary industry-added value, the higher the PCCE. This is roughly consistent with our expectations, as the secondary industry is dominated by high energy-consuming industries, while the tertiary industry has higher

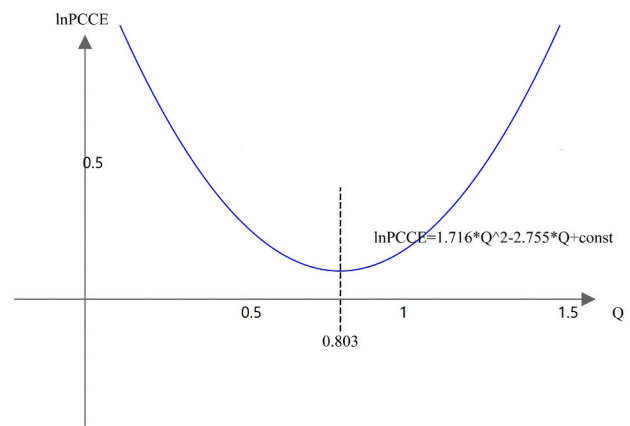


Fig. 8. The U-shaped relationship between polycentricity and PCCE.

added value and lower energy consumption. To some extent, the evolution of industrial structure toward a higher level helps to improve carbon emission performance. (3) The technological progress brought about by R&D investment can improve production efficiency and energy use efficiency, thereby reducing carbon emissions (Wang, Dong, & Dong, 2022).

Considering that the full sample estimation results of the benchmark regression model reflect an average effect, the differences in the development stage of urban agglomerations may affect the role of polycentricity in PCCE. To obtain more accurate estimation results and further validate Hypothesis 2, a heterogeneity analysis was conducted around the above factors (Table 5). Columns I, II, and III illustrate the impact of polycentricity on the PCCE of member cities in the OUUA, DSUA, and NUA, respectively. For the OUUA and DSUA, the coefficient of Q was significantly positive at the statistical level of 1 %, while the NUA met the requirements of the 1 % significance test with the coefficient of Q being -1.516 . Hypothesis 2, which states that urban agglomerations with higher levels of development are more likely to benefit from polycentric development in terms of carbon reduction, is validated.

To test Hypothesis 3, we added the regional transportation infrastructure (RI) variable and its interaction term with polycentricity ($Q \times RI$) to the baseline regression model. The regression results are shown in Table 6. In Column I, the coefficients of Q and $Q \times RI$ are significantly positive, indicating that the establishment of high-speed rail stations significantly enhances the effect of polycentricity on carbon emissions. Since polycentricity is a negative indicator, this means that in OUUA, a higher level of regional transportation infrastructure is beneficial for strengthening the carbon reduction effect of polycentric development. Column III shows the coefficient of Q is significantly negative while that of $Q \times RI$ is significantly positive. In NUA, a higher level of regional transportation infrastructure helps mitigate the carbon emission increase effect of a polycentric spatial structure. Therefore, for less developed urban agglomerations that are not yet subject to agglomeration diseconomies, enhancing regional transportation infrastructure can also help reduce carbon emissions.

To test Hypothesis 4, we introduced the technological innovation

variable and its interaction term with polycentricity. According to Columns IV–VI, for all three types of urban agglomerations, the coefficient of the interaction term is significantly positive. For OUUA and DSUA, the coefficient of Q is also significantly positive, indicating technological innovation enhances the effect of polycentricity on carbon emissions. Therefore, Hypothesis 4 is validated. For highly developed urban agglomerations, stimulating technological innovation is conducive to the release of carbon emission reduction potential in polycentric development. Additionally, for NUA, the coefficient of Q is significantly negative, which is contrary to the sign of $Q \times TECH$. It indicates that technological innovation exhibits a negative moderating effect on less-developed urban agglomerations.

5. Discussion

Compared to this study, some studies conducted in developed regions suggest that the role of polycentric development is generally moderate. This may be related to “the multi-dimensional nature” and “the multi-scale nature” of polycentricity, where different effects overlap and complicate the impact of polycentric development. On the one hand, polycentric spatial structures have a positive impact on the environment (Muñiz & Dominguez, 2020). In the United States, polycentric development in metropolitan areas has created various industrial clusters within urban agglomerations, leading to higher efficiency (Meijers & Burger, 2010). On the other hand, polycentric structures tend to offer lower support for public transportation compared to monocentric urban areas (Lee & Lee, 2014). Some scholars have pointed out that more compositional polycentric forms can lead to increased carbon emissions, while more configurational polycentric forms can reduce carbon emissions (Jung et al., 2022).

Considering the differences in region and scale for urban agglomerations in developing countries, the role of polycentric development in influencing urban efficiency and carbon emissions may become more significant. Even when population sizes are similar, developing and developed countries adapt to population growth in completely different ways. In developing countries, most residents tend to migrate to higher-income cities, gradually leading to urban overcrowding (Wang et al., 2025). This trend suggests that the polycentric development strategy in developing countries could be more effective in mitigating agglomeration inefficiencies and reducing carbon emissions. This study innovatively provides an empirical analysis in the context of Chinese urban agglomerations, contributing to a deeper understanding of the impact of urban agglomeration spatial structure on urban carbon emissions.

The principal discovery of this study highlights a U-shaped relationship between the polycentricity of urban agglomerations and the PCCE of their member cities. In efforts to reduce carbon emissions, excessive concentration in a single centre or an entirely polycentric development with equal resource distribution proves to be ineffective. This finding aligns with previous research, which consistently indicates that the relationship between polycentricity and carbon emissions is not linear (Wang et al., 2020; Wang, Niu, et al., 2022). Unlike linear models, nonlinear models better capture the complex relationship between urban agglomeration development and carbon emissions. Specifically, there is an optimal polycentric structure that leads to minimum carbon emissions. This phenomenon can be attributed to two main factors: (1) When the central city is not large enough to induce significant diseconomies, dispersing the population and industries across multiple centres reduces carbon emission efficiency. Smaller cities, which lack agglomeration benefits, may inefficiently utilize land resources, harming overall energy efficiency (Li et al., 2018). Additionally, establishing a polycentric structure demands substantial investments in municipal infrastructure, which increases energy consumption and could result in higher carbon emissions. (2) A well-organized polycentric network within urban agglomerations can reduce cross-regional and long-distance logistics, thereby improving carbon efficiency. Furthermore, government policies that guide the balanced development of both

Table 5
Regression coefficient of heterogeneity analysis.

Variable	I (OUUA)	II (DSUA)	III (NUA)
Q	0.585*** (0.117)	0.895*** (0.108)	-1.516 *** (0.166)
ECO	0.070*** (0.003)	0.067*** (0.004)	0.052*** (0.006)
POP	-0.185 *** (0.017)	-0.292 *** (0.045)	-1.680 *** (0.249)
IND	-0.039 *** (0.014)	-0.075 *** (0.014)	0.084*** (0.019)
FDI	4.350** (1.844)	-0.673 (2.071)	1.655 (1.969)
GOV	1.371*** (0.087)	2.183*** (0.101)	1.194*** (0.121)
INF	0.027*** (0.002)	0.002 (0.004)	0.028*** (0.004)
$TECH$	-0.012 *** (0.002)	-0.022 ** (0.004)	0.011 (0.011)
HDD	0.005 (0.013)	0.020** (0.010)	0.013 (0.009)
CDD	0.019 (0.013)	0.076*** (0.015)	0.089 (0.056)
_cons	-0.671 *** (0.109)	-0.336 *** (0.073)	2.140*** (0.169)
Individual fixed effect/time fixed effect	YES	YES	YES
R^2	0.658	0.720	0.797
N	1656	1098	558

Remarks: Standard error value is in (); * passed the 10 % significance test; ** passed the 5 % significance test; *** passed the 1 % significance test.

Table 6
Regression coefficient of mechanism analysis.

Variable	I (OUUA)	II (DSUA)	III (NUA)	IV (OUUA)	V (DSUA)	VI (NUA)
Q	0.340** (0.141)	0.768*** (0.129)	−1.841*** (0.170)	0.248* (0.142)	0.739*** (0.115)	−1.830*** (0.178)
RI	−0.040 (0.034)	0.009 (0.032)	−0.404*** (0.073)			
Q × RI	0.108*** (0.035)	0.060 (0.046)	0.515*** (0.083)			
TECH				−0.038*** (0.006)	−0.051*** (0.009)	−0.211*** (0.052)
Q × TECH				0.027*** (0.007)	0.047*** (0.013)	0.247*** (0.057)
_cons	−0.423*** (0.132)	−0.256** (0.085)	2.472*** (0.173)	−0.347*** (0.133)	−0.212*** (0.080)	2.504*** (0.187)
Individual fixed effect/time fixed effect	YES	YES	YES	YES	YES	YES
Control variable	YES	YES	YES	YES	YES	YES
R ²	0.669	0.728	0.813	0.662	0.724	0.804
N	1656	1098	558	1656	1098	558

Remarks: Standard error value is in ().

* passed the 10 % significance test.

** passed the 5 % significance test.

*** passed the 1 % significance test.

population and industry promote equitable land-use practices, mitigating the adverse environmental impacts within the socio-economic system (Wang et al., 2021).

This study also highlights the heterogeneity in the impact of urban agglomeration polycentricity on the PCCE of member cities among urban agglomerations with different levels of development. Urban agglomerations with higher levels of development are more likely to benefit from polycentric development in terms of carbon reduction. The development stage of urban agglomerations is closely related to their spatial structure, implying significant differences in the development level and radiation-driving effect of central cities in different urban agglomerations (Chen, Chen, & Song, 2021).

For urban agglomerations with higher development levels, such as OUUA and DSUA, their central cities are more mature than those in NUA. Larger and more densely populated city centres will suffer from more severe congestion and environmental problems, but they can better connect with sub-centres with stronger radiation (Pan et al., 2024). Therefore, the polycentric development of urban agglomerations can not only alleviate the agglomeration diseconomies caused by central cities but also drive the overall carbon emission efficiency of urban agglomerations through significant radiative driving effects.

Conversely, for urban agglomerations with lower development levels, the central cities tend to be smaller and have weaker agglomeration economies. Decentralized development in these areas may lead to the dispersion of regional resource allocation, making it difficult to form agglomeration economies at the urban level and hindering green and efficient development (Han et al., 2022; He et al., 2021). Thus, the creation of new city centres in such regions should be approached with stricter scrutiny and caution. These findings align with Yu's proposal, which emphasizes the importance of constructing a customized improvement trajectory for each region, taking into account its developmental stage and city scale to optimize spatial structures (Yu, 2021).

Besides the differences in developmental stages, urban agglomerations also exhibit variation in traffic organization layout, technological innovation levels, and other factors. Compared to previous literature, this paper does not merely examine the direct impact of polycentricity on urban PCCE but considers regional transportation infrastructure and technological effects comprehensively. In the mechanism analysis, we found that the establishment of high-speed rail stations has a moderating effect: on the one hand, it enhances the carbon reduction effect of polycentric development in highly developed urban agglomerations, while on the other hand, it alleviates the carbon emission increase effect

of polycentric development in underdeveloped urban agglomerations. Research has shown that high-speed rail can achieve emission reduction goals by decreasing the frequency of use of traditional railways, cars, and other high-emission vehicles through the “substitution effect” (Li & Cheng, 2022). Furthermore, regional transportation infrastructure not only helps optimize and upgrade the internal structure of manufacturing during the polycentric development process, promoting cities toward low-pollution, high-value-added directions, but also fosters the spatial aggregation and development of the service industry, thereby reducing the proportion of heavy industry. High-speed rail has an “indirect crowding-out effect” on industrial structure, manifested as the indirect displacement of high-pollution industries and enterprises (Guo et al., 2021). The traffic organization layout of urban agglomerations may influence the aggregation and dispersion of the spatial distribution of economic activities, thus affecting energy utilization efficiency and carbon emissions through agglomeration effects and economies of scale (Zhang et al., 2020).

The impact of technological innovation on reducing carbon emissions is notably stronger in urban agglomerations with a higher level of development, particularly in polycentric structures. While some studies suggest that technological advancement or the growth of high-tech industries may lead to increased demand for higher-quality products and living standards—thereby fostering the production of short-lived, high-quality goods and frequent product upgrades which can increase carbon emissions (Du et al., 2022)—this study finds that technological innovation contributes to carbon emission reduction. This outcome can be attributed to the efficiency improvements in production processes that stem from technological advancements, ultimately leading to a more pronounced reduction in emissions (Wang, Xie, & Zhang, 2023). In the process of polycentric development in highly developed urban agglomerations, significant knowledge spillover and technology diffusion effects are present. The synergy of both is more conducive to achieving carbon reduction in the member cities of urban agglomerations.

It is noteworthy that this study finds a negative moderating effect of technological innovation on the relationship between polycentricity and carbon emissions in underdeveloped urban agglomerations. When these urban agglomerations promote monocentric development to achieve carbon reduction, technological innovation may diminish the reduction effect. One possible explanation is that, compared to the importation and replicative imitation of technology, original innovation carries much higher costs and risks of failure (Schewe, 1996). Original innovation requires a well-developed system of industry-university-research

collaboration and a concentrated environment of Innovative elements (Caragliu et al., 2016). However, less developed urban agglomerations generally do not meet these conditions, which may lead to inefficient resource utilization due to attempts at original independent innovation (Fan et al., 2021). Some research reveals that R&D activities do not show an electricity-saving effect; indigenous innovation and technical import are complementary when economic development is at a relatively low level, and imitation innovation can improve energy productivity and carbon productivity (Kang et al., 2018). In this context, less developed urban agglomerations should leverage their latecomer advantages, collaborate with highly developed regions and advance technological progress through the importation or imitation of technology.

6. Conclusion and implication

Urban agglomerations typically exhibit high levels of energy consumption and carbon emissions due to their dense populations and intensive economic activities. In the pursuit of carbon neutrality and peak carbon emissions targets, it is crucial to examine the complex relationship between spatial structures and carbon emissions within these agglomerations. However, due to insufficient and often contradictory empirical findings regarding the impact of polycentricity on carbon reduction efforts, this issue remains underexplored. To address this gap, we employed exploratory spatiotemporal data analysis methods to characterize the carbon emission features of 184 prefecture-level cities across 14 urban agglomerations in China and used the polycentricity index to measure the spatial structure of these agglomerations. We then utilized a set of panel fixed-effects models to estimate the effects of urban agglomeration polycentricity on the PCCE of member cities. By comprehensively incorporating multi-dimension control variables and testing moderating effects, this study enhances the accuracy of estimations regarding the influence of polycentricity on PCCE and its underlying mechanisms. Our findings provide valuable empirical evidence to support the low-carbon framework and strategic urban agglomeration planning interventions. The main conclusions are as follows:

- (1) Carbon emissions in urban agglomerations in China show an overall increasing trend in annual variations, with a rapid rise followed by a gradual increase, marked by 2013 as the turning point. Seasonal variations peak in summer and winter.
- (2) The spatial distribution pattern of per capita carbon emissions exhibits a “higher in the north, lower in the south” characteristic, with an enhanced spatial lock-in effect.
- (3) Most urban agglomerations have a polycentric spatial structure, and significant events such as administrative adjustments may alter resource allocation among cities, leading to fluctuations and non-linear changes in polycentricity.
- (4) The relationship between the polycentricity of urban agglomerations and the per capita carbon emissions of member cities follows a U-shaped curve.
- (5) The polycentric development of urban agglomerations does not always guarantee optimal carbon emission performance for member cities in all situations. Urban agglomerations with higher levels of development are more likely to benefit from polycentric development in terms of carbon reduction.
- (6) Improving regional transportation infrastructure and stimulating technological innovation is conducive to unlocking the carbon emission reduction potential in polycentric development.

The findings of this study highlight several key policy recommendations. First, establishing a regional carbon emission coordination mechanism is essential. Given the notable disparities in carbon emission levels between highly developed urban agglomerations such as OUUA and less developed ones like NUA, the government should promote cross-agglomeration collaboration to facilitate balanced and effective

carbon reduction efforts. One viable strategy is the implementation of a cross-regional carbon emissions trading market, which would enhance incentive compatibility for green and low-carbon transitions across various regions. Additionally, urban agglomerations such as HBEY and CS exhibit a High-High cluster pattern of carbon emissions, where spatial spillover effects contribute to the formation of high-emission clusters. In these cases, localized emission reduction measures may be insufficient to counteract the negative externalities generated by adjacent high-emission areas. To address this challenge, a structured consultation mechanism for low-carbon development should be established within urban agglomerations. This mechanism should define overarching carbon reduction targets, facilitate their disaggregation into actionable regional policies, and promote coordinated mitigation strategies.

Second, beyond social and economic considerations, regional emission reduction strategies must also incorporate the impact of spatial structures on carbon emissions. The identified U-shaped relationship emphasizes that fostering a rational and well-structured urban agglomeration pattern is essential for promoting green development. Therefore, it is crucial to establish a scientifically sound spatial framework that aligns with the developmental status of different urban agglomerations. China has continuously optimized and upgraded urban agglomerations, particularly in its eastern regions, which are characterized by high concentrations of population, industry, and resources. While this development has significantly propelled economic growth, it has also introduced challenges, including the widening gap between regional economic expansion and the carrying capacity of natural resources and the environment (Zhou et al., 2018). Policymakers must recognize that urban agglomerations are at different stages of development and implement tailored policy measures accordingly. Highly developed urban agglomerations like OUUA must prioritize the promotion of coordinated development, fostering a polycentric and networked spatial pattern of urban agglomerations. The less developed urban agglomerations like NUA should enhance efficiency, attract resource elements to central cities, and reinforce economies of scale and agglomeration effects.

Third, to further enhance carbon reduction efforts, highly developed urban agglomerations with polycentric development should focus on improving regional transportation infrastructure and stimulating technological innovation. Continuous investment in transportation infrastructure should be pursued to strengthen economic linkages, enhance resource mobility and allocation efficiency, and expand the spatial influence of central cities. A well-integrated transport network can facilitate industrial cooperation, optimize commuting patterns, and reduce reliance on carbon-intensive modes of transportation. Leveraging the synergy of technology is also crucial in maximizing the “innovation effect.” It is important to encourage innovative cooperation between regions, especially those energy-saving and emission-reduction technologies closely related to energy and ecology. On the other hand, for underdeveloped urban agglomerations that are more suited to a monocentric structure, infrastructure improvements and technological innovation currently play a limited role in emission reduction. Therefore, the government can bridge the technological gap by opening up the market and introducing technology. For example, the government should establish industrial cooperation zones to promote direct interaction and connection between cities and obtain better technology based on local conditions.

It is undeniable that this study has certain limitations, warranting further exploration in future research. From a data perspective, the ODIAC dataset used to calculate PCCE in this study represents a territorial emission inventory rather than a consumption-based emissions inventory. Unlike territorial inventories, consumption-based inventories assign emissions to the regions where products are ultimately consumed. Conceptually, consumption-based emissions are derived by subtracting the emissions embodied in exported products from a region's territorial emissions and adding the emissions embedded in imported products

produced elsewhere. However, due to data limitations, accurately accounting for locally produced and consumed goods like cement remains challenging. As a result, the ODIAC dataset cannot fully capture inter-regional carbon flows associated with trade, which is a limitation shared with other existing emission data products. Additionally, urban carbon emissions stem not only from fossil fuel combustion and industrial production but also from land-use changes, which are not comprehensively reflected in the dataset. Future research should prioritize advancing consumption-based carbon accounting methods and refining carbon emission inventories to improve measurement accuracy. From a methodological perspective, this study focuses on the morphological polycentricity of urban agglomerations without considering functional polycentricity. Functional polycentricity—capturing the connections between cities through dimensions such as transportation, economic mobility, and investment networks—provides a more comprehensive understanding of spatial structures. Investigating the influence of functional polycentricity on carbon emissions represents a promising direction for future empirical research, potentially providing deeper insights into the spatial and economic mechanisms shaping carbon efficiency within urban agglomerations.

CRedit authorship contribution statement

Linlin Dai: Writing – review & editing, Project administration, Funding acquisition, Conceptualization. **Jie Luo:** Writing – original draft, Funding acquisition, Formal analysis, Data curation.

Declaration of competing interest

None.

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Data availability

Data will be made available on request.

References

- Ahmad, M., Akram, W., Ikram, M., Shah, A. A., Rehman, A., Chandio, A. A., & Jabeen, G. (2021). Estimating dynamic interactive linkages among urban agglomeration, economic performance, carbon emissions, and health expenditures across developmental disparities. *Sustainable Production and Consumption*, 26, 239–255.
- Akar, I. (2019). The pollution haven hypothesis and foreign trade—a survey on the relation with environment. *Economics Literature*, 1(1), 37–50.
- Bai, X., Blanco, H., Gurney, K. R., Kılış S., Lucon, O., Murakami, J., Pan, J., Sharifi, A., Yamagata, Y., & White, L. (2022). Urban Systems and other settlements. In Intergovernmental Panel on Climate Change (IPCC) (Ed.), *Climate change 2022 - mitigation of climate change* (pp. 861–952). Cambridge University Press.
- Bai, X., Surveyer, A., Elmqvist, T., Gatzweiler, F. W., Güneralp, B., Parnell, S., & Webb, R. (2016). Defining and advancing a systems approach for sustainable cities. *Current Opinion in Environmental Sustainability*, 23, 69–78.
- Brinkman, J. C. (2016). Congestion, agglomeration, and the structure of cities. *Journal of Urban Economics*, 94, 13–31.
- Burgalassi, D., & Luzzati, T. (2015). Urban spatial structure and environmental emissions: A survey of the literature and some empirical evidence for Italian NUTS 3 regions. *Cities*, 49, 134–148.
- Cai, M., Ren, C., Shi, Y., Chen, G., Xie, J., & Ng, E. (2023). Modeling spatiotemporal carbon emissions for two mega-urban regions in China using urban form and panel data analysis. *Science of the Total Environment*, 857, Article 159612.
- Cai, Z., Ding, X., Lin, H., Zhou, Z., Yang, X., & Jiang, P. (2023). Does urban scale borrowing aggravate carbon and air pollution emissions? *Environmental Impact Assessment Review*, 102, Article 107223.
- Caragliu, A., Del Bo, C. F., Kourtit, K., & Nijkamp, P. (2016). The winner takes it all: Forward-looking cities and urban innovation. *The Annals of Regional Science*, 56(3), 617–645.
- Chen, W., Gu, T., Fang, C., & Zeng, J. (2023). Global urban low-carbon transitions: Multiscale relationship between urban land and carbon emissions. *Environmental Impact Assessment Review*, 100, Article 107076.
- Chen, X., Chen, X., & Song, M. (2021). Polycentric agglomeration, market integration and green economic efficiency. *Structural Change and Economic Dynamics*, 59, 185–197.
- Chen, X., Zhang, S., & Ruan, S. M. (2021). Polycentric structure and carbon dioxide emissions: Empirical analysis from provincial data in China. *Journal of Cleaner Production*, 278(12), Article 123411.
- Cirilli, A., & Veneri, P. (2014). Spatial structure and carbon dioxide (CO₂) emissions due to commuting: An analysis of Italian urban areas. *Regional Studies*, 48(12), 1993–2005.
- Dong, J., & Li, C. (2022). Structure characteristics and influencing factors of China's carbon emission spatial correlation network: A study based on the dimension of urban agglomerations. *Science of the Total Environment*, 853, Article 158613.
- Du, L., Jiang, H., Adebayo, T. S., Awosusi, A. A., & Razaq, A. (2022). Asymmetric effects of high-tech industry and renewable energy on consumption-based carbon emissions in MINT countries. *Renewable Energy*, 196, 1269–1280.
- Eeckhout, J., Pinheiro, R., & Schmidheiny, K. (2014). Spatial sorting. *Journal of Political Economy*, 122(3), 554–620.
- Fan, F., Dai, S., Zhang, K., & Ke, H. (2021). Innovation agglomeration and urban hierarchy: Evidence from Chinese cities. *Applied Economics*, 53(54), 6300–6318.
- Fang, C., Bao, C., & Ma, H. (2016). *China urban agglomeration development report 2016* (pp. 17–199). Beijing: Science Press.
- Fang, G., Gao, Z., Tian, L., & Fu, M. (2022). What drives urban carbon emission efficiency? – Spatial analysis based on nighttime light data. *Applied Energy*, 312, Article 118772.
- Friedlingstein, P., O'Sullivan, M., Jones, M. W., Andrew, R. M., Gregor, L., Hauck, J., ... Zheng, B. (2022). Global carbon budget 2022. *Earth System Science Data*, 14(11), 4811–4900.
- Gallego, F., Montero, J.-P., & Salas, C. (2013). The effect of transport policies on car use: Evidence from Latin American cities. *Journal of Public Economics*, 107, 47–62.
- Guo, H., Chen, C., Dong, X., & Jiang, C. (2021). The evolution of transport networks and the regional water environment: The case of Chinese high-speed rail. *Regional Studies*, 55(6), 1084–1110.
- Gurney, K. R., Romero-Lankao, P., Seto, K. C., Hutrya, L. R., Duren, R., Kennedy, C., ... Sperling, J. (2015). Climate change: Track urban emissions on a human scale. *Nature*, 525(7568), 179–181.
- Hall, P. (2009). Looking backward, looking forward: The city region of the mid-21st century. *Regional Studies*, 43(6), 803–817.
- Han, F., Xie, R., & Lu, Y., Fang, J., & Liu, Y. (2018). The effects of urban agglomeration economies on carbon emissions: Evidence from Chinese cities. *Journal of Cleaner Production*, 172, 1096–1110.
- Han, X., Dou, J. M., & Tang, C. H. (2022). Polycentricity, agglomeration, and industrial air pollution in the Chinese City-regions. *Frontiers in Environmental Science*, 10, Article 79395.
- Hao, Y., Chen, H., Wei, Y.-M., & Li, Y.-M. (2016). The influence of climate change on CO₂ (carbon dioxide) emissions: An empirical estimation based on Chinese provincial panel data. *Journal of Cleaner Production*, 131, 667–677.
- He, X., Cao, Y., & Zhou, C. (2021). Evaluation of polycentric spatial structure in the urban agglomeration of the pearl river delta (PRD) based on multi-source big data fusion. *Remote Sensing*, 13(18), 3639.
- Hong, S., Hui, E. C. M., & Lin, Y. (2022). Relationship between urban spatial structure and carbon emissions: A literature review. *Ecological Indicators*, 144, Article 109456.
- Hörcher, D., De Borger, B., Seifu, W., & Graham, D. J. (2020). Public transport provision under agglomeration economies. *Regional Science and Urban Economics*, 81, Article 103503.
- Huang, Y., Hong, T., & Ma, T. (2020). Urban network externalities, agglomeration economies and urban economic growth. *Cities*, 107, Article 102882.
- Hunt, R. A., & Fund, B. R. (2016). Intergenerational fairness and the crowding out effects of well-intended environmental policies. *Journal of Management Studies*, 53(5), 878–910.
- Jung, M. C., Kang, M., & Kim, S. (2022). Does polycentric development produce less transportation carbon emissions? Evidence from urban form identified by night-time lights across US metropolitan areas. *Urban Climate*, 44(14), Article 101223.
- Kang, Z.-Y., Li, K., & Qu, J. (2018). The path of technological progress for China's low-carbon development: Evidence from three urban agglomerations. *Journal of Cleaner Production*, 178, 644–654.
- Lee, S., & Lee, B. (2014). The influence of urban form on GHG emissions in the US household sector. *Energy Policy*, 68, 534–549.
- Li, L., Lei, Y., Wu, S., He, C., Chen, J., & Yan, D. (2018). Impacts of city size change and industrial structure change on CO₂ emissions in Chinese cities. *Journal of Cleaner Production*, 195, 831–838.
- Li, L., Ma, S., Zheng, Y., & Xiao, X. (2022). Integrated regional development: Comparison of urban agglomeration policies in China. *Land Use Policy*, 114, Article 105939.
- Li, X., & Cheng, Z. (2022). Does high-speed rail improve urban carbon emission efficiency in China? *Socio-Economic Planning Sciences*, 84, Article 101308.
- Liu, K., Xue, M. Y., Peng, M. J., & Wang, C. X. (2020). Impact of spatial structure of urban agglomeration on carbon emissions: An analysis of the Shandong peninsula. *China. Technological Forecasting and Social Change*, 161, 9.
- Liu, S., Shen, J., Liu, G., Wu, Y., & Shi, K. (2022). Exploring the effect of urban spatial development pattern on carbon dioxide emissions in China: A socioeconomic density distribution approach based on remotely sensed nighttime light data. *Computers, Environment and Urban Systems*, 96, Article 101847.

- Liu, Z., Guan, D. B., Wei, W., Davis, S. J., Ciais, P., Bai, J., ... He, K. B. (2015). Reduced carbon emission estimates from fossil fuel combustion and cement production in China. *Nature*, 524(7565), 335–.
- Ma, Q., Jia, P., Sun, C., & Kuang, H. (2022). Dynamic evolution trend of comprehensive transportation green efficiency in China: From a spatio-temporal interaction perspective. *Journal of Geographical Sciences*, 32(3), 477–498.
- Mao, B., Gu, J., & Lu, Q. (2023). A pipeline for urban knowledge spillover: Based on the internal linkage of cross-regional multilocation enterprises. *Cities*, 143, Article 104585.
- Meijers, E. J., & Burger, M. J. (2010). Spatial structure and productivity in US metropolitan areas. *Environment and Planning A*, 42(6), 1383–1402.
- Muñiz, I., & Dominguez, A. (2020). The impact of urban form and spatial structure on per capita carbon footprint in US larger metropolitan areas. *Sustainability*, 12(1), 19, Article 389.
- Oda, T., Maksyutov, S., & Andres, R. J. (2018). The open-source data inventory for anthropogenic CO₂, version 2016 (ODIAC2016): A global monthly fossil fuel CO₂ gridded emissions data product for tracer transport simulations and surface flux inversions. *Earth System Science Data*, 10(1), 87–107.
- Pan, H., Yao, Y., Ming, Y., Hong, Z., & Hewings, G. (2024). Whither less is more? Understanding the contextual and configurational conditions of polycentricity to improve urban agglomeration efficiency. *Cities*, 149, Article 104884.
- Phelps, N. A., Miao, J. T., & Zhang, X. (2023). Polycentric urbanization as enclave urbanization: A research agenda with illustrations from the Yangtze River Delta region (YRDR). *China. Territory Politics Governance*, 11(2), 261–280.
- Qian, Y., Wang, H., & Wu, J. S. (2022). Spatiotemporal association of carbon dioxide emissions in China's urban agglomerations. *Journal of Environmental Management*, 323, 12.
- Rey, S. J., & Janikas, M. V. (2009). STARS: Space-time analysis of regional systems. In *Handbook of applied spatial analysis: Software tools, methods and applications* (pp. 91–112). Springer.
- Rocky Mountain Institute. (2019). Carbon emission peaking path of China's City clusters: Exploring the opportunities. <https://www.efchina.org/Reports-en/report-lccp-20190804-en>.
- Sat, N. A. (2018). Polycentricity in a developing world: A micro-regional analysis for morphological polycentricity in Turkey. *GeoScape*, 12(2), 64–75.
- Schewe, G. (1996). Imitation as a strategic option for external acquisition of technology. *Journal of Engineering and Technology Management*, 13(1), 55–82.
- Shen, W., Liang, H., Dong, L., Ren, J., & Wang, G. (2021). Synergistic CO₂ reduction effects in Chinese urban agglomerations: Perspectives from social network analysis. *Science of the Total Environment*, 798, Article 149352.
- Solis, E., Karimi, K., Garcia, I., & Mohino, I. (2022). Knowledge economy clustering at the Intrametropolitan level: Evidence from Madrid. *Journal of the Knowledge Economy*, 13 (2).
- Tabuchi, T. (1998). Urban agglomeration and dispersion: A synthesis of Alonso and Krugman. *Journal of Urban Economics*, 44(3), 333–351.
- The State Council of China. (2021). Notice of the State Council on Issuing the Action Plan for Peaking Carbon Emissions Before 2030. https://www.gov.cn/zhengce/zhengeku/2021-10/26/content_5644984.htm.
- Veneri, P., & Burgalassi, D. (2012). Questioning polycentric development and its effects. Issues of definition and measurement for the Italian NUTS-2 regions. *European Planning Studies*, 20(6), 1017–1037.
- Volgmann, K., & Rusche, K. (2020). The geography of borrowing size: Exploring spatial distributions for German urban regions. *Tijdschrift voor Economische en Sociale Geografie*, 111(1), 60–79.
- Wang, C., Wang, L., Xue, Y., & Li, R. (2022). Revealing spatial spillover effect in high-tech industry agglomeration from a high-skilled labor flow network perspective. *Journal of Systems Science and Complexity*, 35(3), 839–859.
- Wang, C., Zhan, J., Zhang, F., Liu, W., & Twumasi-Ankrah, M. J. (2021). Analysis of urban carbon balance based on land use dynamics in the Beijing-Tianjin-Hebei region. *China. Journal of Cleaner Production*, 281, Article 125138.
- Wang, F., Fan, W., Liu, J., Wang, G., & Chai, W. (2020). The effect of urbanization and spatial agglomeration on carbon emissions in urban agglomeration. *Environmental Science and Pollution Research*, 27, 24329–24341.
- Wang, F., Rani, T., & Razzaq, A. (2023). Environmental impact of fiscal decentralization, green technology innovation and institution's efficiency in developed countries using advance panel modelling. *Energy & Environment*, 34(4), 1006–1030.
- Wang, J., Dong, X., & Dong, K. (2022). How does ICT agglomeration affect carbon emissions? The case of Yangtze River Delta urban agglomeration in China. *Energy Economics*, 111, Article 106107.
- Wang, M., Li, Y., & Liao, G. (2022). Spatial spillover and interaction between high-tech industrial agglomeration and urban ecological efficiency [original research]. *Frontiers in Environmental Science*, 10.
- Wang, S. J., & Li, M. Y. (2023). Impact of spatial structure of urban agglomerations on PM_{2.5} pollution: Based on resource misallocation. *Heliyon*, 9(3), 13.
- Wang, Y., Sun, B., & Zhang, T. (2022). Do polycentric urban regions promote functional spillovers and economic performance? Evidence from China. *Regional Studies*, 56(1), 63–74.
- Wang, Y. N., Niu, Y. J., Li, M., Yu, Q. Y., & Chen, W. (2022). Spatial structure and carbon emission of urban agglomerations: Spatiotemporal characteristics and driving forces. *Sustainable Cities and Society*, 78, 10.
- Wang, Y. S., Shi, G. L., & Zhang, Y. B. (2025). How did polycentric spatial structure affect carbon emissions of the construction industry? A case study of 10 Chinese urban clusters. *Engineering Construction and Architectural Management*, 32(2), 1186–1210.
- Wang, Z. S., Xie, W. C., & Zhang, C. Y. (2023). Towards COP26 targets: Characteristics and influencing factors of spatial correlation network structure on US carbon emission. *Resources Policy*, 81, 14.
- Xie, R., Fang, J., & Liu, C. (2017). The effects of transportation infrastructure on urban carbon emissions. *Applied Energy*, 196, 199–207.
- Yan, D. S., & Li, P. X. (2023). Can regional integration reduce urban carbon emission? An empirical study based on the Yangtze River Delta, China. *International Journal of Environmental Research and Public Health*, 20(2), 25.
- Yang, H., Lu, Z., Shi, X., Muhammad, S., & Cao, Y. (2021). How well has economic strategy changed CO₂ emissions? Evidence from China's largest emission province. *Science of the Total Environment*, 774, Article 146575.
- Yang, J. T., Lei, Y., Su, H., & Yu, K. H. (2022). Evaluation and exploration on the implementation of policy-oriented construction method of urban agglomeration spatial structure: Based on the development process of Zhongyuan urban agglomeration. *Urban Development Studies*, 29(03), 70–79.
- Yang, L., Wang, Y. Q., Lian, Y. J., & Han, S. S. (2020). Factors and scenario analysis of transport carbon dioxide emissions in rapidly-developing cities. Transportation research part D-transport and environment, 80, 27, article 102252.
- Yu, B. (2021). Urban spatial structure and total-factor energy efficiency in Chinese provinces. *Ecological Indicators*, 126, Article 107662.
- Yu, X., Wu, Z. Y., Zheng, H. R., Li, M. Q., & Tan, T. L. (2020). How urban agglomeration improve the emission efficiency? A spatial econometric analysis of the Yangtze River Delta urban agglomeration in China. *Journal of Environmental Management*, 260, 8.
- Yu, Y., Wei, Q., Li, M., & Yi, L. (2020). Spatial optimization strategies of population function in China's world-class urban agglomerations during 14th five-year plan period. *Bulletin of Chinese Academy of Sciences (Chinese Version)*, 35(7), 835–843.
- Zhang, K. Y., & Zhang, J. (2022). Polycentricity and green development efficiency of urban agglomerations. *China Population, Resources and Environment*, 32(2), 107–117.
- Zhang, P., Zhao, Y., Zhu, X., Cai, Z., Xu, J., & Shi, S. (2020). Spatial structure of urban agglomeration under the impact of high-speed railway construction: Based on the social network analysis. *Sustainable Cities and Society*, 62, Article 102404.
- Zhou, D., Bonafoni, S., Zhang, L., & Wang, R. (2018). Remote sensing of the urban heat island effect in a highly populated urban agglomeration area in East China. *Science of the Total Environment*, 628, 415–429.
- Zhou, K., Yin, Y., Li, H., & Shen, Y. (2021). Driving factors and spatiotemporal effects of environmental stress in urban agglomeration: Evidence from the Beijing-Tianjin-Hebei region of China. *Journal of Geographical Sciences*, 31(1), 91–110.
- Zhu, X., Wang, Q., Zhang, P., Yu, Y., & Xie, L. (2021). Optimizing the spatial structure of urban agglomeration: Based on social network analysis. *Quality & Quantity*, 55, 683–705.
- Zhuo, Y. X., & Liu, T. (2020). Polycentric city and region: A review and appraisal. *Progress in Geography*, 39(08), 1385–1396.
- Zou, Y. F., Lu, Y. H., & Cheng, Y. (2019). The impact of polycentric development on regional gap of energy efficiency: A Chinese provincial perspective. *Journal of Cleaner Production*, 224, 838–851.